Ventilator Hyperinflation Procedure

1. Scope

Site	Operational Area	Applicable to
Armadale Health Service	Intensive Care Unit	Physiotherapy

2. Purpose

Ventilator Hyperinflation (VHI) is a physiotherapy intervention that enables the delivery of larger than baseline tidal volumes (VT) via adjustment to the ventilator in the intubated and ventilated patient.

VHI has been studied as a treatment mode by Physiotherapists for lung recruitment in the presence of atelectasis and to facilitate the mobilisation of secretions in cases of sputum retention. Both Manual Hyperinflation (MHI) and VHI have been demonstrated to be effective in the treatment and management of sputum retention.

Prior to performing the technique in the Intensive Care Unit (ICU), physiotherapists must complete a competency assessment conducted with a Senior Physiotherapist who works within the ICU.

3. Aim of VHI

To improve the respiratory status of intubated, ventilated patients by the delivery of larger than normal tidal volume breaths, utilising safe peak inspiratory pressure without interruption to the desired positive end expiratory pressure (PEEP) and the oxygen supply.

VHI enables recruitment of collapsed alveoli and clearance of secretions from the bronchi, thereby improving gas exchange. VHI is implemented as a treatment option by Physiotherapists only after a full respiratory assessment to determine the individual patient needs.

3.1 Indications

VHI is seen as an adjunct to other Physiotherapy treatments, such as positioning and manual techniques. VHI is indicated in the stable ventilated patient to:

- Improve lung compliance
- Mobilise and remove excessive bronchial secretions
- Aid in the resolution of atelectasis
- Improve impaired gas exchange
- Assist with prevention of nosocomial pneumonia

Any concerns about performing VHI on a patient should be discussed with the Senior Physiotherapist, Senior Medical Officer and or Senior Nursing Staff prior to commencing VHI.

3.2 Contraindications:

3.2.1 Absolute Contraindications

3.2.1.1 Pathological Conditions:

- Severe bronchospasm or gas trapping in asthma
- Haemoptysis, active lung or tracheostomy bleeding
- End-stage pulmonary fibrosis
- Increase intracranial pressure on neuroprotective parameters / acute undiagnosed head trauma
- Severe coagulopathy.
- Documented cystic lung changes (bullae / blebs) such as in moderate to severe chronic obstructive pulmonary disease (COPD) with large emphysematous bullae or cavitating lung pathology
- Subcutaneous emphysema
- Undrained pneumothorax or intercostal catheter with an air leak
- Obstructing airway tumour or lung tumour at risk of cavitation
- Untreated or massive pulmonary embolus
- Bronchopleural fistula
- Patients requiring nitric oxide or prostacyclin infusion

3.2.1.2 Cardiovascular Parameters:

- Mean arterial blood pressure (MAP) < 65mmHg or > 100mmHg except in a head injured patient where a higher MAP is used to maintain cerebral perfusion pressure (CPP)
- Unstable heart rhythm
- Inotropic requirement equivalent of >15 ml / hour (8mg in 100ml) of noradrenaline or a sudden increase in inotropes
- Post insertion of central venous catheter / PICC line prior to chest x-ray (CXR) clearance

3.2.1.3 Respiratory Parameters:

- Airway Pressure Release Ventilation (APRV) mode of ventilation
- High PEEP > 10cm H_2O
- Fractional inspired oxygen (FiO_2) > 0.70
- Peak Airway Pressure (P_{peak}) > 30cm H_2O
- Intrinsic PEEP > 5
- Unexplained increase in respiratory rate

3.2.2 Relative Contraindications:

- Level of consciousness for patient compliance
- Recent thoracic surgery (diaphragmatic repair, pneumonectomy, oesophagectomy, lung volume reduction surgery (LVRS)
- Rib fractures, flail segments and suspected pneumothorax
- Pneumothorax with patent intercostal catheter (ICC)
- Pneumomediastinum

- Noradrenaline running between 10-15ml /hour (8mg in 100ml). This requires a discussion with the consultant before proceeding
- Multiple inotropic agents i.e. Noradrenaline, Dobutamine, Adrenaline or anti-arrhythmic medications
- Acute Respiratory Distress Syndrome (ARDS)

All identified relative contraindications must be discussed with an ICU Consultant before initiating treatment.

4. Procedure

4.1 Procedure

- Patient must be assessed for suitability and need for VHI
- Position patient optimally to optimise sputum drainage during VHI
- Check for any contraindications
- Discuss the procedure with the Primary Nurse or Senior Nurse on duty
- Explain the procedure to the patient if appropriate
- Apply personal protective equipment as per infection control guidelines
- Determine target VT by calculating the patients BMI using Appendix A: BMI Weight Chart and Appendix B: Estimating Height from Ulnar Length

$$\text{BMI} = \frac{\text{Weight (kg)}}{\text{Height}^2\ (\text{m}^2)}$$

- If BMI is ≤ 25 determine target VT at 15ml / kg based on their weight
- If BMI is > 25 determine target VT at 15ml / kg based on their ideal body weight for their height with BMI of 25

Take note of the following pre intervention:

- Ventilator mode settings
- Alarm Parameters
- VT
- EtCo2
- P_{peak}

4.1.1 Set up procedure for Hamilton C6

- Press “Modes”
- Select “DuoPAP” mode
- Press “Confirm”
- FiO_2 and PEEP Low (P_L) will automatically transfer and match existing pre-set levels
- Change P_{peak} alarm to 40cm H_2O
- Adjust VT alarm to target VT plus 300ml
- Reduce respiratory rate “Rate” to 8 / min and confirm by pressing it again
- Set “Time High” (T_H) between 3-5 seconds and confirm by pressing it again
- Set “PEEP High” (P_H) at 20cm H_2O and confirm by pressing it again. Press “Confirm” to start VHI in DuoPAP mode.

- Increase P_H incrementally by 2cm H_2O until target volume is achieved.
- Settings will depend on patient's lung compliance, pre-existing conditions or precautions
- Adjust "Rate" to maintain baseline Minute Ventilation (ExpMinVol on Hamilton C6)
- If P_{Peak} increases too high, increase Time High (T_H) (up to 5 seconds if not already done) and if necessary, decrease "Rate" (minimum 6 / min)
- Aim for 3 sets of 10 breaths at target volume. Perform first set then reassess.
- Continue onto second and third if patient continues to meet requirements as above. Include chest wall vibrations if indicated.
- Return to pre-treatment parameters (ventilation mode, VT, Rate, FiO_2, T_H, P_{Peak} and VT alarms)
- Suction as required following pre-oxygenation
- Document settings used and outcome measures
- Cross check with bedside Nurse that all parameters have been reset to pre-treatment level

4.1.2 Potential Negative Responses to VHI:

- If Peak airway pressures increase > 30cm H_2O when establishing larger VT:
- increase T_H up to 5 seconds and if necessary
- decrease Rate (minimum 6 / min)
- Patient may become stressed during the intervention (tachycardia, hypertension); if observed reassure patient and consider bolus sedation
- Reduce VT if necessary, to see if VHI is better tolerated, proceed with slower incremental rises in VT, modify T_H and f to achieve better synchrony
- Accept VT at a lower level than target VT and reassess tolerance in subsequent sets of VHI
- VHI may increase intra-thoracic pressure, reducing venous return and cardiac output. It is important to recognise haemodynamic instability prior to commencing treatment and to be vigilant during intervention, ceasing if instability is noted
- If there is marked deterioration in SpO_2/HR/BP discontinue VHI and assess for reasons it has occurred
- When applying VHI in the patients on neuroprotective parameters, monitor VT and $EtCO_2$ and ensure they are closely matching pre-treatment levels (fluctuating levels and inadequate ventilation can influence ICP)

4.2 Monitoring During VHI

Throughout performing VHI, it is important to monitor the following:

- Blood pressure (BP)
- Heart rate (HR)
- Oxygen saturations (SpO_2)
- End tidal CO_2 ($EtCO_2$) / Minute Ventilation (ExpMinVol)
- Signs of patient distress
- Peak airway pressure (P_{peak})
- Plateau pressures

5. Documentation

Document the following in the patient progress notes as per AKG Documentation Standards – Clinical Procedure:

- BMI and target VT
- Patient position
- Number of breaths delivered
- P_H, P_L and T_H
- Maximum volumes reached
- Patients' response to treatment:
    - Plateau pressures
    - Gas exchange (arterial blood gas (ABG) if appropriate or available)
    - SpO2
    - Sputum clearance
    - CXR / auscultation findings
    - Wave forms (expiratory flow waveform / morphology)
    - ICP / EtTCO2 where indicated
    - Haemodynamic
- Plan for frequency and dosage of treatment

6. Roles and responsibilities

The ICU physiotherapist will review the relevant guidelines which will be endorsed by the Head of Department Intensive Care Unit and HOD Physiotherapist. All physiotherapists must complete a supervised training program and competency-based assessment (see Appendix B) with a senior ICU physiotherapist before unsupervised practice is allowed. The physiotherapist, medical team and nursing staff must be in partnership for the delivery of the intervention.

7. Supporting information

The following documents support the implementation of this procedure:

- Intubation and Management of the Ventilated Patient in the Intensive Care Unit
- Documentation Standards – Clinical Procedure
- Recognition and Response Acute Deterioration Policy

8. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards:

- Standards 1

  - Action 1.27: The health service organisation has processes that provide clinicians with ready access to best-practice guidelines, integrated care pathways, clinical pathways and decision support tools, relevant to their clinical practice. Support clinicians to use the best available evidence, including relevant clinical care standards developed by the Australian Commission on Safety and Quality in Health Care
- Standard 3
  - Action 3.12: The health service organisation has processes for the appropriate use and management of invasive medical devices that are consistent with the current edition of the Australian Guidelines for the prevention and Control of Infection in Healthcare
- Standard 5
  - Action 5.11: Clinicians comprehensively assess the conditions and risks identified through the screening process
  - Action 5.14: The comprehensive care plan is used to direct the delivery of safe and effective care that aligns with the patient's needs and preferences
- Standard 8
  - Action 8.04: The health service organisation has processes for clinicians to detect acute physiological deterioration that require clinicians to:
    - Document individualised vital sign monitoring plans
    - Monitor patients as required by their individualised monitoring plan
    - Graphically document and track changes in agreed observations to detect acute deterioration over time, as appropriate for the patient

10. References

1. Berney, S., & Denehy, L. (2002). A comparison of the effects of manual and ventilator hyperinflation on static lung compliance and sputum production in intubated and ventilated intensive care patients. *Physiotherapy Research International, 7*(2), 100-108. http://dx.doi.org/10.1002/pri.246
2. Savian, C., Paratz, J., & Davies, A. (2005) Comparison of the effectiveness of manual and ventilator hyperinflation at different levels of PEEP. *National Cardiothoracic Group 9th Biennial Conference, 52*(1) Supplement, S25.
3. Dennis, D. M., Jacob, W.J., & Budgeon, C. (2012). Ventilator versus manual hyperinflation in clearing sputum in ventilated intensive care patients. *Anaesthesia and Intensive Care, 40*(1), 142-149.

11. Glossary

The following definitions are relevant to this policy as seen on Hamilton C6:

Term	**Definition**
VT	Tidal Volume
ExpMinVol	Minute Ventilation
f total	Respiratory rate
Rate	Frequency
Ppeak	Peak Airway Pressure
Th	Time High

Term	Definition
P_H	Pressure High
P_L	Pressure Low

12. Policy owner (relating to these procedures)

Head of Department Physiotherapy

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13. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3.0	08/09/2025	08/09/2028	Minor amendments

14. Authorisation

This procedure has been authorised and issued by the Director Clinical Services.

Authorised by	Danielle Kilmurray, Director Allied Health & Ambulatory Care
Authorisation date	12 September 2025
Published date	01 October 2025

Appendix A: BMI Weight Chart

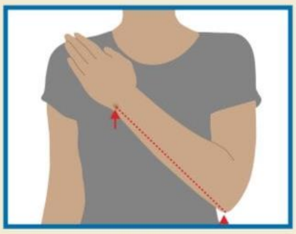
BMI Weight Chart - Centimetres/Kilograms - (BMI: 19 - 35)

1. First select your height then select your weight.
2. Select the nearest value/s to your own if they are not displayed in the chart.
3. Your Body Mass Index will be listed at the top and bottom of the BMI chart.

	Normal Weight (18.5 - 24.9)						Overweight (25 - 29.9)					Obesity (=> 30)					
BMI	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
	Body Weight - (kilograms)																
Height - cms (metres)																	
147cm (1.47m)	41	44	45	48	50	52	54	56	59	61	63	65	67	69	72	73	76
150cm (1.50m)	43	45	47	49	52	54	56	58	60	63	65	67	69	72	74	76	78
152cm (1.52m)	44	46	49	51	54	56	58	60	63	65	67	69	72	74	76	79	81
155cm (1.55m)	45	48	50	53	55	57	60	62	65	67	69	72	74	77	79	82	84
157cm (1.57m)	47	49	52	54	57	59	62	64	67	69	72	74	77	79	82	84	87
160cm (1.60m)	49	51	54	56	59	61	64	66	69	72	74	77	79	82	84	87	89
163cm (1.63m)	50	53	55	58	61	64	66	68	71	74	77	79	82	84	87	89	93
165cm (1.65m)	52	54	57	60	63	65	68	71	73	76	79	82	84	87	90	93	95
168cm (1.68m)	54	56	59	62	64	67	70	73	76	78	81	84	87	90	93	95	98
170cm (1.70m)	55	57	61	64	66	69	72	75	78	81	84	87	90	93	96	98	101
172cm (1.72m)	57	59	63	65	68	72	74	78	80	83	86	89	92	95	98	101	104
175cm (1.75m)	58	61	64	68	70	73	77	80	83	86	89	92	95	98	101	104	107
178cm (1.78m)	60	63	66	69	73	76	79	82	85	88	92	95	98	101	104	107	110
180cm (1.80m)	62	65	68	71	75	78	81	84	88	91	94	98	101	104	107	110	113
183cm (1.83m)	64	67	70	73	77	80	83	87	90	93	97	100	103	107	110	113	117
185cm (1.85m)	65	68	72	75	79	83	86	89	93	96	99	103	107	110	113	117	120
188cm (1.88m)	67	70	74	78	81	84	88	92	95	99	102	106	109	113	116	120	123
191cm (1.91m)	69	73	76	80	83	87	91	94	98	102	105	109	112	116	120	123	127
193cm (1.93m)	71	74	78	82	86	89	93	97	100	104	108	112	115	119	123	127	130

Appendix B: Estimating Height from Ulnar Length

Estimating height from ulna length



Measure between the point of the elbow (olecranon process) and the midpoint of the prominent bone of the wrist (styloid process) (left side if possible).

Height (m)	men (<65 years)	1.94	1.93	1.91	1.89	1.87	1.85	1.84	1.82	1.80	1.78	1.76	1.75	1.73	1.71
	men (≥65 years)	1.87	1.86	1.84	1.82	1.81	1.79	1.78	1.76	1.75	1.73	1.71	1.70	1.68	1.67
	Ulna length (cm)	32.0	31.5	31.0	30.5	30.0	29.5	29.0	28.5	28.0	27.5	27.0	26.5	26.0	25.5
Height (m)	Women (<65 years)	1.84	1.83	1.81	1.80	1.79	1.77	1.76	1.75	1.73	1.72	1.70	1.69	1.68	1.66
	Women (≥65 years)	1.84	1.83	1.81	1.79	1.78	1.76	1.75	1.73	1.71	1.70	1.68	1.66	1.65	1.63

Height (m)	men (<65 years)	1.69	1.67	1.66	1.64	1.62	1.60	1.58	1.57	1.55	1.53	1.51	1.49	1.48	1.46
	men (≥65 years)	1.65	1.63	1.62	1.60	1.59	1.57	1.56	1.54	1.52	1.51	1.49	1.48	1.46	1.45
	Ulna length (cm)	25.0	24.5	24.0	23.5	23.0	22.5	22.0	21.5	21.0	20.5	20.0	19.5	19.0	18.5
Height (m)	Women (<65 years)	1.65	1.63	1.62	1.61	1.59	1.58	1.56	1.55	1.54	1.52	1.51	1.50	1.48	1.47
	Women (≥65 years)	1.61	1.60	1.58	1.56	1.55	1.53	1.52	1.50	1.48	1.47	1.45	1.44	1.42	1.40